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RESEARCH ARTICLE

Luc P. J. Selen · Peter J. Beek · Jaap H. van Dieën

Impedance is modulated to meet accuracy demands during goal-directed arm movements

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Abstract The neuromuscular system is inherently noisy and joint impedance may serve to filter this noise. In the present experiment, we investigated whether individuals modulate joint impedance to meet spatial accuracy demands. Twelve subjects were instructed to make rapid, time constrained, elbow extensions to three differently sized targets. Some trials (20 out of 140 for each target, randomly assigned) were perturbed mechanically at 75% of movement amplitude. Inertia, damping and stiffness were estimated from the torque and angle deviation signal using a forward simulation and optimization routine. Increases in endpoint accuracy were not always reflected in a decrease in trajectory variability. Only in the final quarter of the trajectory the variability decreased as target width decreased. Stiffness estimates increased significantly with accuracy constraints. Damping estimates only increased for perturbations that were initially directed against the movement direction. We concluded that joint impedance modulation is one of the strategies used by the neuromuscular system to generate accurate movements, at least during the final part of the movement.

Keywords Precision · Neuromotor noise · Stiffness · Motor variability

Introduction

Signal-dependent neuromotor noise is supposed to underlie variability in biological movement (Schmidt et al. 1979; Harris and Wolpert 1998; Jones et al. 2002; Todorov and Jordan 2002). Because of the noise in neuromuscular transmission and the orderly recruitment of motor units according to the size principle (Jones et al. 2002; Selen et al. 2005), muscular forces show an approximately linear relationship between their mean and standard deviation (SD) (Schmidt et al. 1979; Christou et al. 2002; Todorov 2002). Obviously, such noise will limit the accuracy of goal-directed movements. Historically, research on goal-directed movements has focused on the relation between movement speed and endpoint accuracy. Since the pioneering studies of Woodworth (1899) and Fitts (1954), various studies have corroborated the general finding that movement speed and endpoint accuracy are inversely related (see Plamondon and Alimi (1997) for a review) and this relation has been attributed to signal-dependent neuromotor noise (Schmidt et al. 1979; Harris and Wolpert 1998; Elliott et al. 2001). Nevertheless, the same movement, i.e. same amplitude and same movement time, can be achieved with different levels of endpoint accuracy (Laursen et al. 1998; Gribble et al. 2003; Osu et al. 2004). This raises the question which mechanisms, besides speed, are employed by the motor system to meet spatial accuracy demands.

Van Galen and De Jong (1995) proposed that modulation of joint impedance[1] might be used to control spatial accuracy. To date, joint impedance has been studied mainly in relation to external perturbations (e.g. Burdet et al. 2001; Franklin et al. 2003). The general finding is that increasing joint impedance, both through co-contraction and reflex modulation, stabilizes the limb to external force fields. Neuromuscular noise, however, is an internal source of perturbation. In this case, the muscles, paradoxically, would both form the source of motor variability and provide the means for suppressing its kinematic consequences. Modeling studies have shown that in spite of this paradox, co-activation of muscles can reduce the effects of force variability on kinematics (Van Galen and de Jong 1995; Selen et al. 2005). It is,

L. P. J. Selen · P. J. Beek · J. H. van Dieën (✉)
Faculty of Human Movement Sciences, Institute for Fundamental and Clinical Human Movement Sciences, Vrije Universiteit,
Van der Boechorststraat 9, 1081 BT Amsterdam, The Netherlands
E-mail: j.vandieen@fbw.vu.nl
Tel.: +31-20-5988501
Fax: +31-20-5988529

[1]Throughout this paper impedance refers to the combined effect of stiffness and damping. Otherwise stiffness $(dM/d\varphi)$ *or damping* $(dM/d\dot{\varphi})$ are mentioned explicitly.

however, insufficiently clear whether humans actually use this control strategy.

Indirect measures of increased impedance, like pen pressure and increased EMG amplitudes, have been reported in response to increased accuracy demands (e.g. Van Gemmert and Van Galen 1997; Laursen et al. 1998; Van Galen and Van Huygevoort 2000; Gribble et al. 2003; Osu et al. 2004; Visser et al. 2004; Sandfeld and Jensen 2005; Van Roon et al. 2005). Although muscular co-activation and stiffness are related (Osu et al. 2002), direct estimates of stiffness and damping are required to quantify the magnitude of impedance modulation and to account for the dissociation of EMG and impedance with fatigue (Zhang and Rymer 2001).

Thus far, to our knowledge, the relation between kinematic variability and impedance was only examined in two studies, in both cases as a corollary of the main research question. Shiller et al. (2002) reported that the kinematic variability in vowel production co-varied with jaw stiffness. However, the stiffness was largely determined by the jaw geometry in this study, which suggests that the reported co-variation was merely a by-product of the geometry rather than an active motor control strategy. Furthermore, accuracy was not manipulated and thus it is unknown whether the impedance level was modulated to match accuracy demands. The experiment of Burdet et al. (2001) provided more convincing evidence for the usefulness of stiffness regulation with accuracy demands. They invited subjects to make point-to-point movements in a negative elastic force field perpendicular to the movement direction. In order to hit the target, subjects increased the stiffness of the arm selectively in the direction of the force field. Strikingly, trajectory variability was lower after removal of the force field than prior to exposure to the force field, indicating that stiffness modulation can indeed help to diminish kinematic variability.

Although there are some indications that impedance might be modulated in response to accuracy demands, direct mechanical evidence is lacking. The present study aims at filling this lacuna. Endpoint accuracy demands were manipulated in time-constrained elbow movements. Estimates of elbow impedance were obtained by applying torque perturbations to the arm during movement. We hypothesised that subjects modulate joint impedance to meet accuracy demands.

Methods

Subjects

Twelve subjects (four males and eight females) between 20 and 28 years of age participated in the experiment. All subjects had normal or corrected to normal vision and no (history of) neuromuscular disorders. The local Ethics Committee approved the experiment and all subjects signed informed consent forms prior to the experiment. The experiment took less than 4 h and subjects were allowed to rest frequently to avoid fatigue.

Apparatus

Subjects were seated on a chair in front of a semicircular array of light emitting diodes (LEDs). Their preferred forearm, including the hand palm and wrist, was cast (NobaCast, Noba Verbandmittel Danz GmbH) onto a lightweight T-wedged bar. The bar was mounted onto the vertical motor shaft of a torque controlled motor (S-motor, elu93028, Fokker Control Systems), with the medial epicondyle aligned with the motor axis of rotation and the palm of the hand facing downwards.

The chair height was adjusted such that the upper arm and forearm were in the horizontal plane. The upper arm was in line with the shoulders. The LED-array, consisting of 447 LEDs, was placed 1.5 m in front of the wrist of the cast arm. The arm pointed to the centre of the LED-array at an elbow angle of 90°. A small laser pointer was attached to the lightweight bar and indicated the pointing direction on the LED-array. Four LEDs were illuminated, defining the boundaries of the start and target areas (see Fig. 1).

The torque controlled motor operated in closed loop at 5 kHz. In the unperturbed trials the set point of the controller was 0 Nm, allowing smooth and frictionless movements by the subject. The angular position of the motor shaft was measured by a potentiometer (22HSPP-10, Sakae) and the remaining torque was measured by a strain gauge. Both position and torque were stored at 1 kHz.

In the perturbed trials, the set point changed when passing the 75% point of the movement amplitude. This point was chosen because in an earlier study (Osu et al. 2004) both kinematic variability and muscular activity were influenced by accuracy demands only in the final part of the movement, suggesting that joint impedance was controlled only, or at least predominantly, in the final part of the movement. The applied torque pulse had a duration of 140 ms. The torque pulse changed sign after 70 ms in order to prevent the optimization routine from getting trapped by the co-linearity of angle, angular velocity and angular acceleration. Torque amplitude was set to 5 Nm. The total motor-subject dynamics prohibited the system from exactly generating this value (see Fig. 2), but torque profiles were reproducible within and between experimental conditions.

Experimental task

Subjects were asked to perform rapid pointing movements, 0.26 rad in 300 ms, by elbow extension from the start area to the target area. Three blocks, each with a differently sized target, of 165 trials were executed. Blocks, with small, medium and large target areas, corresponding to 0.015, 0.030 and 0.045 rad elbow angle,

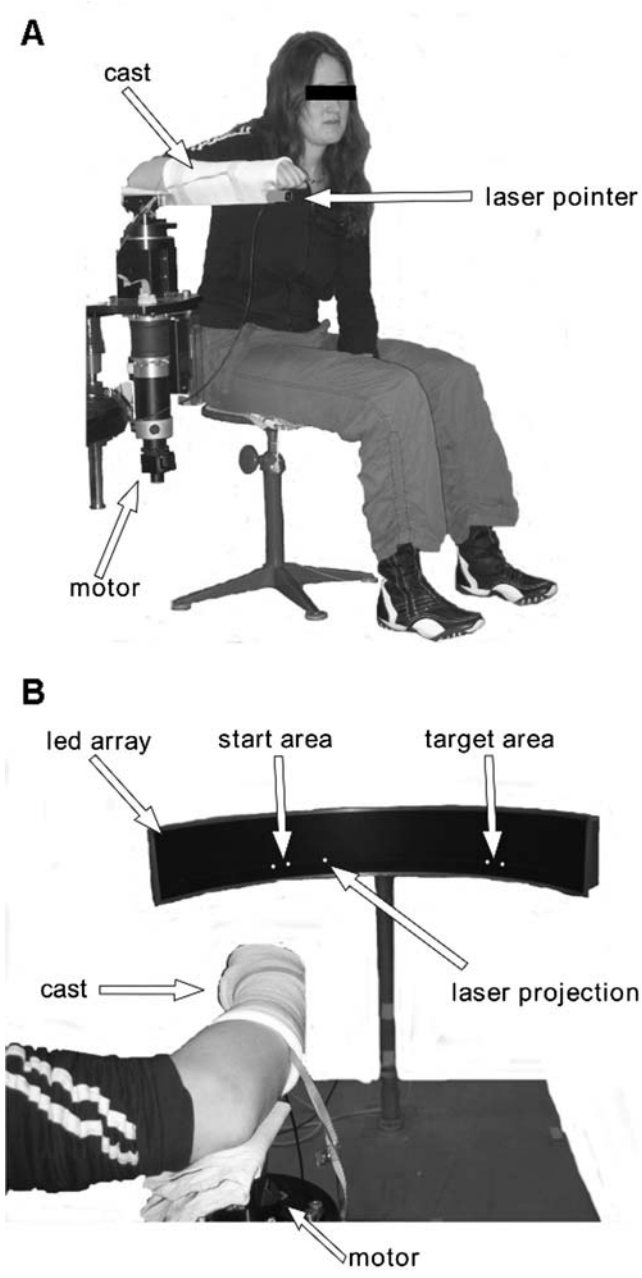

Fig. 1 Experimental setup for the goal-directed movements with different target areas. **a** Side-front view of a subject with the forearm cast onto a lightweight bar attached to the motor. **b** The led array with the start area, target area and the laser projection. The screen that provided the subject feedback about target based movement time (tarMT) and 'hit' or 'no-hit' after each trial is not depicted

were presented in random order. In all conditions, the distance between the target centres was 0.26 rad. Each block started with 25 practice trials. From the subsequent 140 trials, 20 randomly selected trials were perturbed. Two types of perturbation were applied: (1) initial perturbation in flexion direction, while extending the elbow (pFLEX) and (2) initial perturbation in extension direction, while extending the elbow (pEXT). Figure 2 shows an example of both types of perturbation and their

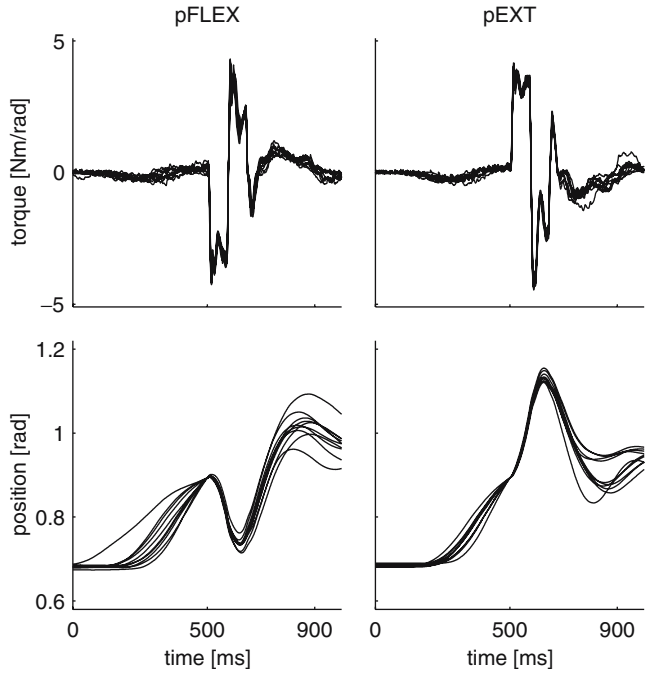

Fig. 2 Examples of the torque and position profiles for the two perturbation types in one subject. pFLEX torque initially opposes movement for 70 ms and subsequently assists the movement for 70 ms. pEXT torque initially assists movement for 70 ms and subsequently opposes movement for 70 ms. Torque amplitude was set to 5 Nm, but total motor-subject dynamics prohibited exact generation of this profile

kinematic consequences. Although the perturbations were applied towards the end of movement, leaving insufficient time (< 170 ms) to voluntarily react to the perturbation before movement completion, subjects were instructed not to intervene with the perturbations and to focus on the next trial to come.

A trial started by illuminating the boundaries of the start and target areas upon which the subject positioned the pointer in the start area. After 500 ms, a short beep indicated that the subject was allowed to start the movement. After 3000 ms, the boundaries of the start and target areas were extinguished. Subsequently, the movement time was calculated and presented to the subject on a computer screen. Movement time was defined as the time between leaving the largest start area and entering the largest target area in order to keep the amplitude over which MT was calculated constant. In addition to the calculated MT, the range of desired movement times (270–330 ms) was presented. The inclusion of a MT criterion was intended to suppress the natural strategy to move slower when confronted with higher accuracy constraints. Furthermore, a 'hit' or 'no-hit' signal was presented to the participant in the form of a green or red button on the same screen as the MT. A 'hit' required the subject to stay within the target area for at least 500 ms after entering it. For every temporally as well as spatially correct trial the participant earned five Eurocents. In order to keep the subject motivated, the total accumulated credit was also presented on the screen.

Prior to the actual experiment, the same perturbations, seven times within a time-span of 15 s, were applied to the stationary arm in neutral position, i.e. 90° elbow flexion in two conditions. In the 'relaxed' condition, subjects were instructed not to tense their muscles and not to respond to the perturbations. From these perturbations independent estimates of combined arm and manipulandum inertia were obtained. In the 'stiff' condition, subjects were instructed to maximally co-activate their muscles and to minimize the angular displacements as a result of the perturbations. The calculated values for stiffness and damping were used to quantify relative stiffness and damping during movement.

Effectiveness of accuracy manipulation and MT feedback

The effectiveness of the accuracy manipulation and the MT feedback was investigated in both the entire dataset and in the subset of trials in which the imposed temporal and the spatial constraints were matched (see above). In the present study, we looked for strategies other than speed to meet accuracy demands. Therefore MT should not systematically change with target size. Temporal changes were investigated for two measures of movement time. The first measure was the one presented to the subject; it was defined as the time between leaving and entering the largest target area (from now on called 'target based movement times': (tarMT). The second measure was the time between movement start and movement end (from now on called ' kinematics based movement times': (kinMT). Movement start was defined as the instant at which the angular velocity first exceeded 0.05 rad/s. Movement end was defined as the first instant at which the pointer was within the largest target area and the angular velocity was below 0.05 rad/s. Furthermore, the maximum velocity and the time to peak velocity after movement onset were computed in order to check for changes in the velocity profiles.

Spatial accuracy in relation to target size was investigated by calculating the SD of the position at the end of the movement. Furthermore, we counted all trials, for each target area, in which the spatial and temporal constraints for the smallest target area were matched. If the accuracy manipulation had been effective, the number of trials thus counted should decrease with larger targets.

Trajectory variability was assessed by calculating the between trial SD as a function of time. Before calculating the evolution of the SD with time, position data were time normalised, between movement onset and movement end, by cubic spline interpolation.

Estimation of elbow impedance

The dynamics of a joint in response to an external mechanical perturbation can be approximated by a *K-B-I* model (K = stiffness, B = damping and I = inertia):

$$M_{\mathrm{ext}} = K \cdot \varphi_{\mathrm{rel}} + B \cdot \dot{\varphi}_{\mathrm{rel}} + I \cdot \ddot{\varphi}_{\mathrm{rel}}. \tag{1}$$

The external moment (M_{ext}) leads to a position (φ_{rel}), velocity ($\dot{\varphi}_{\mathrm{rel}}$) and acceleration ($\ddot{\varphi}_{\mathrm{rel}}$) change relative to the intended unperturbed trajectory. The intended propagation of the perturbed trajectory can only be estimated by extrapolation of the trajectory prior to perturbation (Burdet et al. 2000; Popescu et al. 2003). Extrapolation is based on selection of unperturbed trajectories that are similar, in a least squares sense, to the perturbed trajectory prior to perturbation. In order to guarantee convergence of the perturbed and unperturbed trajectory towards the instant of perturbation, the distance was weighted exponentially prior to averaging. The similarity (S) between two trajectories was expressed as:

$$S = \sum_{t=t_0-500}^{t=t_0} \left\{ \mathrm{e}^{(t-t_0+500)/w} \cdot \left(\varphi_u(t) - \varphi_p(t)\right)^2 \right\}. \tag{2}$$

The weighting exponential was scaled by a factor w with a value of 100 ms, resulting in a 150-fold weighting of the distance between the trajectories at perturbation relative to the weighting 500 ms prior to perturbation. The variables φ_u and φ_p are the unperturbed and the perturbed position, respectively. Time is expressed in milliseconds in the above equation, with t_0 representing the time that the perturbation angle is traversed.

If for each perturbed trajectory the most similar unperturbed trajectory is chosen for comparison, the number of trials to derive impedance estimates from is equal to the number of perturbed trials. Bearing in mind the imposed spatial and temporal constraints were not met in a significant number of unperturbed trials, it was expected that a similar percentage of the perturbed trials would also not have met the constraints (see Fig. 6). The best possible estimator for the correctness of a perturbed trial is the number of correct unperturbed trials that is similar to the perturbed one up to the point of perturbation. However, analysis of the unperturbed trials revealed that high similarity before a certain point in time does not necessarily result in high similarity after that point in time. Therefore the angular differences between the 10% most similar perturbed and unperturbed trajectories were selected for further analysis (thereby including the same perturbed trial multiple times, but each time with another unperturbed match). The average angular difference between the selected trajectories (φ_{rel}) was taken as input to the parameter optimization procedure.

In order to estimate stiffness, damping and inertia for all experimental conditions, seven optimizations were performed per subject. In the first optimization, the inertia was determined separately, from the perturbations of the relaxed arm. Given the model parameter I and the time series of M_{ext}, the angular response was simulated over 70 ms using Heun's method:

$$\varphi_{\mathrm{sim}}(t) = \int_{t=0}^{t=70\,\mathrm{ms}} \left(\int_{t=1}^{t=70\,\mathrm{ms}} \frac{M_{\mathrm{ext}}(t)}{I} \mathrm{d}t \right) \mathrm{d}t. \tag{3}$$

The inertia was optimized by minimizing the objective function:

$$E_{\text{inertia}}(\varphi_{\text{rel}}, \varphi_{\text{sim}}) = [\varphi_{\text{rel}} - \varphi_{\text{sim}}] \cdot [\varphi_{\text{rel}} - \varphi_{\text{sim}}]^T + [\min(\varphi_{\text{rel}} - \varphi_{\text{sim}}, 0)] \cdot [\min(\varphi_{\text{rel}} - \varphi_{\text{sim}}, 0)]^T \quad (4)$$

The second term in Eq. 4 was introduced to force the optimization to favour solutions that yielded an angular deviation slightly larger than the experimental data. Test optimizations revealed that inertia was overestimated without inclusion of this term, resulting in negative damping values to accommodate for the large inertia in the remaining simulations.

Given the inertia, K and B were determined for the three target areas and the two perturbation types separately over the first 170 ms after the onset of the perturbation. As stated before, within this time window, subjects are unable to voluntarily react to the perturbation due to neuromuscular delays. The objective function to be minimised by optimizing K and B was:

$$E(\varphi_{\text{rel}}, \varphi_{\text{sim}}) = [\varphi_{\text{rel}} - \varphi_{\text{sim}}] \cdot [\varphi_{\text{rel}} - \varphi_{\text{sim}}]^T. \quad (5)$$

Using

$$\ddot{\varphi}_{\text{sim}}(t) = \frac{-B \cdot \dot{\varphi}_{\text{sim}}(t) - K \cdot \varphi_{\text{sim}}(t) + M_{\text{ext}}(t)}{I}, \quad (6)$$

to calculate $\varphi_{sim}(t)$ by numerical integration.

To optimize estimates for I, B and K a simulated annealing algorithm (Goffe et al. 1994) was implemented. After optimization the variance accounted for (VAF) was calculated as an index of the validity of the parameter estimates:

$$\text{VAF} = 1 - \frac{[\varphi_{\text{rel}} - \varphi_{\text{sim}}] \cdot [\varphi_{\text{rel}} - \varphi_{\text{sim}}]^T}{[\varphi_{\text{rel}}] \cdot [\varphi_{\text{rel}}]^T}. \quad (7)$$

A value of 1 indicates that no differences exist between the measured and the simulated position deviation.

Using a similar approach, estimates of maximum stiffness and damping were calculated from the perturbations in the 'stiff' condition.

Statistics

Both in the text and in the figures, the data are presented as means and SD. To examine the effects of target size on movement time and accuracy a repeated measures ANOVA was carried out. Changes in the distribution of MT with target size were investigated using repeated measures MANOVA with the 15th, 50th and 85th percentile of MT as dependent measures. The velocity profiles were checked for changes in skewness using a repeated measures MANOVA with peak velocity and time to peak velocity as dependent measures.

Impedance changes with target size were first examined in a two-way repeated measures MANOVA (Target Size × Perturbation Type) with K and B as measures. The same procedure was applied for the perturbation types separately. Both the one- and two-way MANOVAs were followed by repeated measures ANOVAs for K and B separately. Honest significant difference (HSD) Tukey's tests were used to further analyse significant effects in all cases. An alpha level of 0.05 was chosen for all tests.

Results

Temporal and spatial constraints

The average tarMT for the small, medium and large target area were 300 (SD 31), 291 (SD 17) and 290 (SD

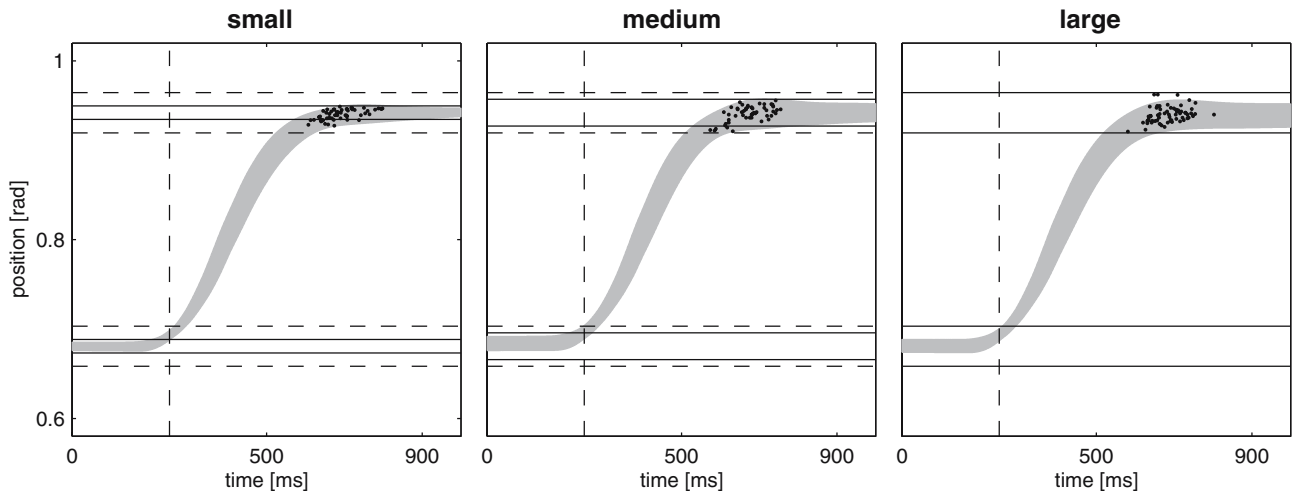

Fig. 3 Kinematic profiles of the correct trials of one subject. Results for the small, medium and large target area are shown from *left* to *right*. The *grey area* indicates the 95% confidence interval of the trials. *Continuous lines* depict the presented target areas and the *dashed lines* the largest target area. Those boundaries were used to calculate tarMT. *Dots* depict the time and position of movement end for the individual trials. All trials were aligned at movement onset, which is indicated by the *vertical dashed line*

15) ms, respectively, and were not significantly affected by the size of the target area [$F_{(2,22)} = 1.285$, $p = 0.297$]. Also the kinMT were not significantly different [$F_{(2,22)} = 0.536$, $p = 0.593$], with values of 513 (SD 49), 508 (SD 27) and 504 (SD 31)ms from the smallest to the largest target area. The variability of tarMT, expressed as the 15th and 85th percentile, differed from the predefined range of 270–330 ms. The 15th percentile was smaller than 270 ms [233 (SD 21)ms, $t_{(35)} = -10.551$, $p < 0.000$] and the 85th percentile was larger than 330 ms [360 (SD 39) ms; $t_{(35)} = 4.534$, $p < 0.000$]. However, there was no effect of target size on the distribution of tarMT [$F_{(6,42)} = 0.487$, $p = 0.814$] nor of kinMT [$F_{(6,42)} = 1,645$, $p = 0.159$] with the 15th, 50th and 85th percentile as measures.

Only those trials that met the tarMT criterion of 270–330 ms were included for further analysis. For this subset, the distribution of tarMT [$F_{(6,42)}$, $p = 0.241$] and kinMT [$F_{(6,42)}$, $p = 0.203$] was also not significantly different between the three target areas. From these trials, only those that remained within the target area for at least 500 ms were selected. After this elimination procedure only 38 (SD 10), 48 (SD 5) and 53 (SD 5)% of the 120 unperturbed trials remained, reflecting the decreasing difficulty with increasing target area [$F_{(2,22)} = 21.573$, $p < 0.000$; see Fig. 4, panel *D*]. However, for this subset kinMT differed significantly between target areas [$F_{(6,42)} = 2.248$, $p = 0.031$], which was attributable to a decreasing kinMT with increasing target area of the 85th percentile of kinMT [$F_{(1.425,15.672)} = 6.126$, $p = 0.017$, see Table 1]. No significant changes were found in the skewness of the velocity profiles as a function of target size [$F_{(42,4)} = 1.173$, $p = 0.336$], with peak velocity and time to peak velocity as measures. The average maximum velocity and time to peak velocity over all subjects and targets were 1.01 (SD 0.05)rad/s and 178 (SD 19) ms, respectively.

Figure 3 shows the 95% confidence interval of the kinematic profiles of the selected trials for the different target areas for a single subject. The spatial dispersion of the endpoints, as indicated by the black dots, increased from the smallest to the largest target area. The same was true for all subjects [$F_{(2,22)} = 36.037$, $p < 0.000$; see Fig. 4a]. Also for the entire dataset, with no constraints on MT, a similar but weaker effect was found [$F_{(2,22)} = 6.086$, $p = 0.008$; see Fig. 4b]. To further confirm that the accuracy demands manipulation was effective, all trials that met the tarMT criterion were analysed as if they had been performed on the smallest target. Trials were approved when they entered the smallest target area and remained there for at least 500 ms. The number of correct trials decreased with increasing target area [$F_{(2,22)} = 6.46$, $p = 0.006$; see Fig. 4c], indicating that subjects made use of the larger areas and purposely suppressed endpoint variability for the smallest target area.

Having established that accuracy increased with smaller target areas, the question remains whether this accuracy was only achieved near the endpoint or that

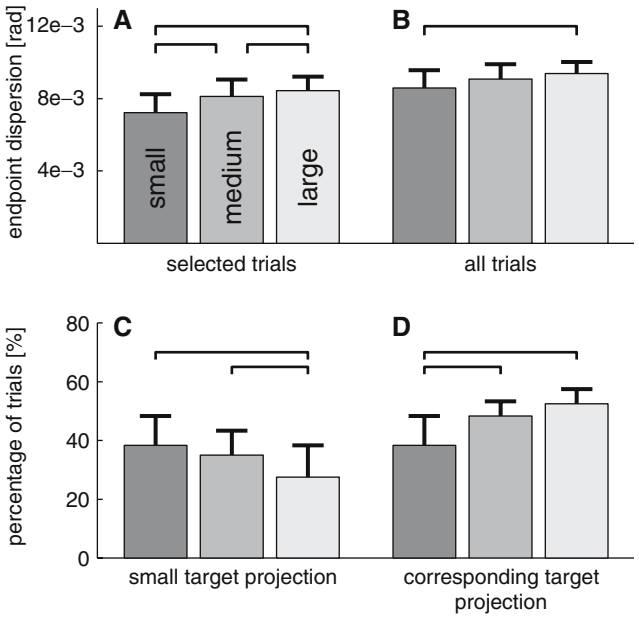

Fig. 4 Means and SD for the small, medium and large target area, for endpoint dispersion (**a**, **b**) and the number of correct trials (**c**, **d**). Endpoint dispersion was expressed as the standard deviation of the movement endpoints, for the selected trials (**a**) and for all trials (**b**). On the lower row the means and SD of the percentage of trials that met the tarMT and 'hit'-criterion for the smallest target area (**c**) and the percentage of correct trials that met the tarMT and 'hit'-criterion for the corresponding target area (**d**) are presented. *Horizontal lines* indicate significant differences between target areas ($p < 0.05$). For **a** S < M, $p = 0.006$; S < L, $p = 0.001$; M < L, $p < 0.000$; **b** S < L, $p = 0.015$; **c** S > L, $p = 0.041$; M > L, $p = 0.013$; **d** S < M, $p = 0.015$; S < L, $p < 0.000$.

overall trajectory variability decreased. As can be appreciated from Fig. 5, for some subjects the variability was lower for the entire trajectory (Fig. 5a), whereas other subjects reduced variability near the target (Fig. 5b). For all subjects, the trajectory variability assumed the order of the target areas after 75% of the trajectory had been traversed (Fig. 5c).

Impedance modulation

Prior to the target size experiment, inertia of the relaxed arm and manipulandum together and stiffness and damping estimates during maximal co-contraction were obtained. The inertia ranged from 0.0435 to 0.0828 Nms^2/rad between subjects. Stiffness and damping during maximal co-contraction ranged from 24.4 to 99.5 Nm/rad and 0.18 to 1.39 Nms/rad, respectively.

Figure 6 shows the 95% confidence interval for unperturbed correct trials of one subject for the medium target area with the perturbed trials superimposed. Notice that only four and five out of ten perturbed trials, for pEXT and pFLEX, respectively, fell within the 95% confidence interval. This was a general finding for all subjects and in accordance with the small number of unperturbed trials in which both temporal and spatial constraints were met. The similarity was calculated

Table 1 Movement times for the small, medium and large target of the selected trials

	KinMT (ms)			TarMT (ms)		
	p15	p50	p85	p15	p50	p85
Small	434 (45)	499 (62)	589 (60)	273 (8)	296 (10)	323 (8)
Medium	430 (23)	485 (33)	563 (38)	270 (4)	292 (6)	320 (8)
Large	431 (31)	481 (36)	549 (43)	269 (3)	291 (6)	319 (5)

The 15th, 50th and 85th percentile (p15, p50 and p85) are presented for both the kinematics based target time (*kinMT*) and the target based movement time (*tarMT*). The only significant effect with target size was found for the 85th percentile of *kinMT*

between all perturbed trials, whether in or outside the 95% confidence interval, and all correct unperturbed trials. The average difference between the 10% most similar trajectories was taken for further analysis. The maximum deviation from the unperturbed trajectory decreased significantly with smaller target areas [$F_{(2,22)} = 8.931$, $p < 0.001$] and interacted with perturbation direction [$F_{(2,22)} = 9.494$, $p = 0.001$]. Post hoc ANOVA's revealed that this interaction was attributable to pFLEX [$F_{(2,22)} = 23.350$, $p < 0.000$], whereas the deviation for pEXT did not change systematically [$F_{(2,22)} = 2.009$, $p = 0.158$]. As will become apparent in the following, a lack of changes in the deviation for pEXT does not necessarily mean that there are no changes in the impedance, because differences in the timing of the peak deviation and in the external torque are still possible.

The dynamics of the elbow joint was well described by the *K-B-I* model with VAFs ranging from 0.9913 to 0.9990. Figure 7 shows the mean and SD for all subjects of the stiffness and damping estimates for the three target areas and the two perturbation types. The repeated measures MANOVA revealed only significant effects for target size and its interaction with perturbation type. The effect for target area was attributable to the stiffness, whereas the interaction effect was attributable to the damping. Post hoc tests for the perturbation types separately revealed that stiffness decreased with increasing target area for both perturbation types, whereas the damping only decreased with increasing target area for pFLEX. Table 2 provides an overview of all test results.

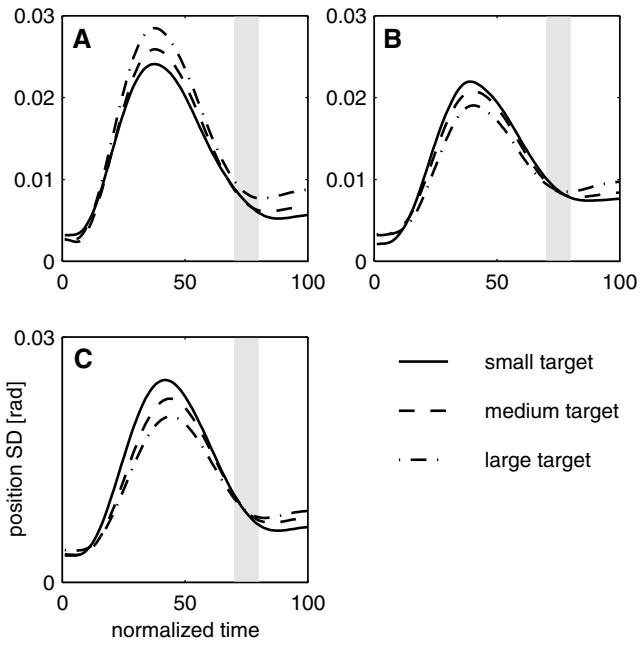

Fig. 5 Position variability as a function of normalized time and target size. **a** Single subject showing smaller variability over the whole trajectory to the smaller target area. **b** Single subject showing higher variability in the first three-fourth of the trajectory to the smaller target area, only meeting the task constraints in the last one-fourth of the trajectory. **c** Averaged time courses across all subjects. *Grey area* indicates the perturbation area, coinciding with the crossing of the traces

Discussion

Impedance modulation with accuracy constraints

We examined the modulation of mechanical impedance, quantified by *K* and *B*, as a function of accuracy demands in goal-directed movements. Mechanical impedance was determined when 75% of the movement amplitude was traversed. It was found that mechanical impedance, probably both of intrinsic and reflexive nature, increased with higher accuracy demands. This is in agreement with earlier studies that reported increased muscular co-activation with increasing accuracy demands, both during single (Osu et al. 2004) and multi-joint (Laursen et al. 1998; Gribble et al. 2003; Visser et al. 2004; Sandfeld and Jensen 2005; Van Roon et al. 2005) movement. Although the estimation of impedance (especially stiffness) changes from EMG might be feasible (Osu et al. 2002), direct estimation of mechanical impedance by means of perturbations is preferable, especially when the relation between EMG and stiffness becomes unreliable, for example because of fatigue (Zhang and Rymer 2001).

It remains an unresolved issue why the positional variability took the order of the accuracy demands only in the final quarter of movement. Similar to our study, Osu et al. (2004) reported that the positional variability was structured only after 80% of movement amplitude had been traversed. In the same study, increases in the muscular activity of the individual muscles in response to increased accuracy demands were only found in the final stage of the movement. This might indicate that positional variability is only controlled, by impedance

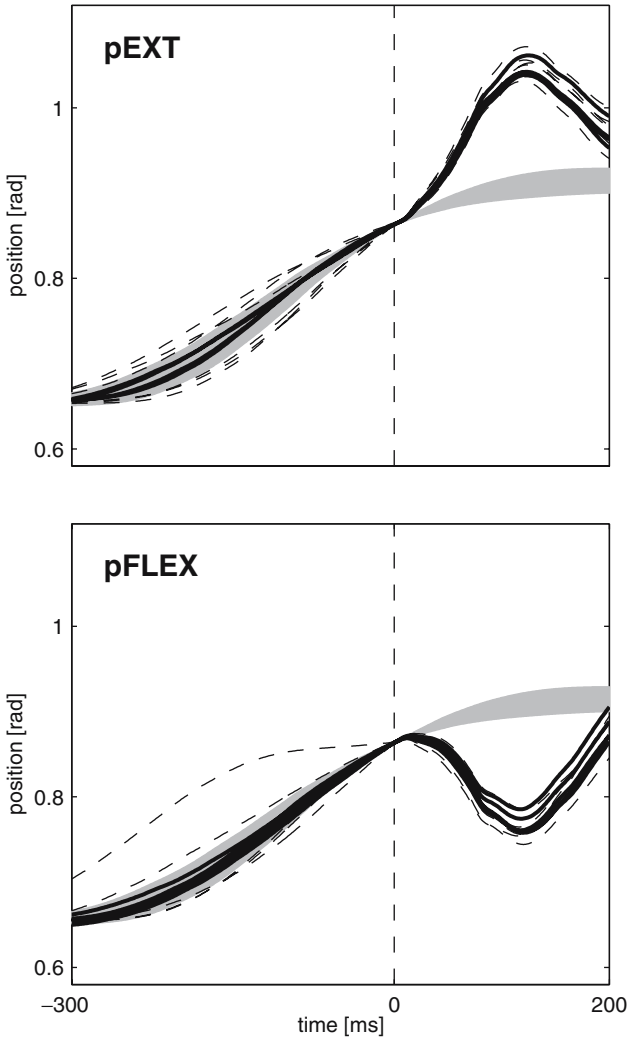

Fig. 6 The 95% confidence interval for the correct unperturbed trials (*grey area*) with superimposed the ten perturbed trials for pEXT and pFLEX for one subject. *Continuous lines* represent the trials that fell within the 95% confidence interval and *dashed lines* those outside this interval. All trials were aligned at the perturbation angle, which is indicated by the *dashed vertical line*

modulation, during this stage. These results are consistent with the suggestion of Todorov and Jordan (2002) that optimal feedback control under signal-dependent neuromotor noise would imply postponing all goal-directed corrections to the last possible moment. Although in their model stiffness is solely of a 'reflexive' nature and does not incorporate muscle impedance, the basic idea fits with our experimental findings.

Parameter values for inertia, damping and stiffness

The inertia values of the forearm, including lightweight cast and apparatus inertia, are in the same range as those reported in the literature (Bennett et al. 1992; Bennett 1993; Popescu et al. 2003). Also the stiffness values are generally comparable to those reported in the literature. For goal-directed movements at constant speed, elbow

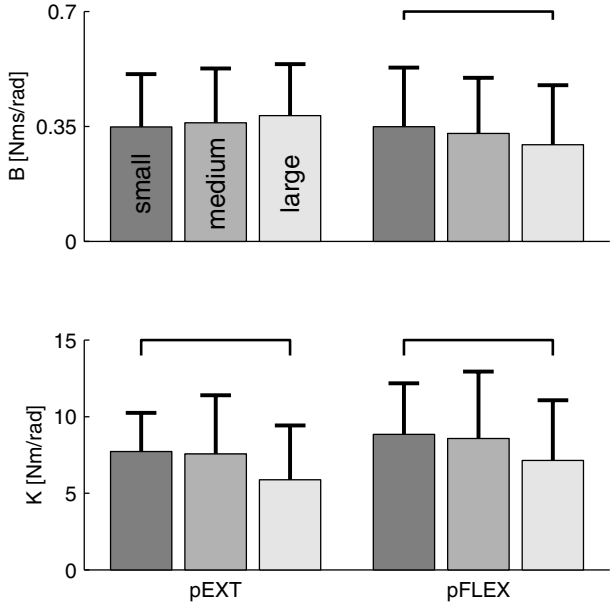

Fig. 7 Mean and SD of damping (*B*) and stiffness (*K*) for all subjects for the two perturbation types as a function of target area. *Horizontal lines* indicate significant differences between target areas ($p < 0.05$). For: (*B*; pFLEX) S > L, $p = 0.014$; (*K*;pEXT) S > L, $p = 0.040$; (*K*,pFLEX) S > L, $p = 0.011$

stiffness values between 5 and 12 Nm/rad have been reported (Kalveram et al. 2005). For goal-directed movement at different speeds, Bennett (1993) reported elbow stiffness values ranging from 3 to 15 Nm/rad depending on net torque changes with speed. Although in our study net torque did not vary, our stiffness estimates were in the same range, probably as a result of muscular co-activation with accuracy demand (Osu et al. 2004) resulting in more coupled cross-bridges and higher muscle stiffness. During cyclic movement, elbow stiffness values ranged from 2 to 15 Nm/rad over the trajectory (Bennett et al. 1992). Only the elbow stiffness values reported by Popescu et al. (2003) during goal-directed movement were five times larger than in the present study. We suspect that this difference was caused by the relatively short (biphasic, 30 ms pulse) and large (20 Nm) perturbations in Popescu's study, resulting in a strong contribution of short-range stiffness to the total muscle stiffness.

Damping values during movement have been reported less frequently in the literature. During cyclic movement the damping of the elbow joint was reported to range from 0 to 0.7 Nms/rad within a cycle (Bennett et al. 1992), whereas the damping during goal-directed movement ranged from 0.4 to 1.2 Nms/rad between subjects (Popescu et al. 2003; Kalveram et al. 2005). Our estimates fell within the same range.

Taking a closer look at the impedance, we observe that, with increasing accuracy demands, the increases in *K* were more prominent than the increases in *B*. When we calculate the damping and stiffness relative to their maximum values, the median relative stiffness is around 10% with peaks towards 40% whereas the relative damping covers the whole range from 5 to 90%,

Table 2 Statistical tests of the effects of Target Size and Perturbation Direction (type: pFLEX and pEXT) on stiffness (*K*) and damping (*B*)

Target size	$F_{(4,44)} = 3.430$	$p = 0.008$*	*K*	$F_{(2,22)} = 6.1689$	$p = 0.011$*
			B	$F_{(2,22)} = 0.009$	$p = 0.651$
Perturbation direction	$F_{(2,10)} = 1.383$	$p = 0.295$	*K*	$F_{(1,11)} = 2.958$	$p = 0.113$
			B	$F_{(1,11)} = 0.963$	$p = 0.348$
Target size x perturbation direction	$F_{(4,44)} = 4.857$	$p = 0.003$*	*K*	$F_{(2,22)} = 0.103$	$p = 0.849$
			B	$F_{(2,22)} = 7.862$	$p = 0.005$*
Target size (pFLEX)	$F_{(4,44)} = 6.422$	$p = 0.001$*	*K*	$F_{(2,22)} = 4.197$	$p = 0.029$*
			B	$F_{(2,22)} = 4.676$	$p = 0.020$*
Target size (pEXT)	$F_{(4,44)} = 2.412$	$p = 0.063$	*K*	$F_{(2,22)} = 5.428$	$p = 0.012$*
			B	$F_{(2,22)} = 2.146$	$p = 0.141$

*Indicates significant result ($p < 0.05$)

suggesting that the damping is controlled to a lesser extent than stiffness. In isometric conditions, stiffness and damping are strongly coupled and increase linearly with activation level (e.g. Hajian and Howe 1997; Zhang and Rymer 1997). However, during goal-directed movement the coupling between stiffness and damping weakens in the stabilization phase (Milner and Cloutier 1998) and during cyclic movement the damping varies much more within and between cycles than the stiffness (Bennett et al. 1992). Also during goal-directed movement, stiffness and damping do not follow the same pattern (Bennett 1994). Although it is not clear why the coupling between stiffness and damping disappears during movement, one might speculate that the strong nonlinearity of the force–velocity relation and the contribution of reflexes are involved.

The methods applied in the present study cannot distinguish between the contributions of intrinsic muscle properties and spinal reflexes to the overall joint impedance. To do so, continuous (pseudo-)random perturbations are necessary. Since the motor control system is adaptive, these perturbations would become part of the motor task and thus interfere with the impedance control in order to suppress the effects of neuromotor noise on kinematics. However, studies on position control have shown, using continuous perturbations in combination with sophisticated system identification techniques, that both intrinsic properties and reflexes contribute simultaneously to joint impedance (Kearney et al. 1997; Zhang and Rymer, 1997; Van der Helm et al. 2002). In all likelihood, this was also the case in our experiment. However, the reflexive contribution was probably only elicited by the perturbation. During unperturbed movement stability was probably guaranteed by intrinsic properties only.

Other control strategies to meet accuracy demands

Changes in impedance (Fig. 7) were less clear-cut than changes in spatial accuracy (Fig. 4a). This difference may be explained in part by methodological factors. For example, the spatial accuracy was calculated only for the spatially and temporally correct trials, whereas it remains uncertain whether the perturbed trials would have been correct. Furthermore, the medium target may already have required maximum performance, but with more hits than when aiming at the small target.

Besides methodological factors this difference may also reflect additional control strategies. It has been proposed that changes in movement variability can be decomposed in three components (Müller and Sternad 2004): (1) reduction of stochastic noise; (2) exploitation of task tolerance; (3) co-variation between central variables. In the present experiment task, tolerance was manipulated both by the presentation of differently sized targets and the constraints on MT. As a result the impedance increased, which is essentially a means to reduce the effects of stochastic noise on motor performance. From Fig. 4 one can appreciate that subjects made use of the tolerance, i.e. target size, offered by the task. Not only impedance was modulated in the present study, traces of speed modulation were still apparent. Within subjects, tarMT fluctuated from −20 to +20% of the desired MT, indicating that the relative contribution of the two strategies, impedance and speed modulation, was not constant. Furthermore, there was a general tendency to move slower with smaller targets, but this came only to the fore in the 85th percentile of the correct trials.

It can only be speculated that the co-variation of central variables was modulated in the present task. The only degrees of freedom available to co-vary are the muscular activations, which we did not assess. However, during learning of multi-joint frisbee throwing co-variance between central variables changed with practice and the accompanying accuracy (Yang and Scholz 2005). Following the uncontrolled manifold concept (UCM concept, Scholz and Schöner 1999), total joint configuration variance was divided in the amount of kinematic variance in the subspace of joint configurations that did not interfere with performance (goal-equivalent variance; GEV) and the amount of variance in its orthogonal subspace in which variability had consequences on task performance (nongoal-equivalent variance; NGEV). Although overall variability decreased with practice, GEV decreased to a larger extent than NGEV. Using the same concept, Kang et al. (2004) showed that during learning of an unusual multi-finger force production task, overall performance increased by selectively (re-)distributing the force variability over the individual fingers. It is unclear whether humans employ the co-variation between central variables when confronted with different accuracy demands, but it is most likely that in natural tasks impedance modulation, speed modulation and variability distribution are employed in

concordance, tailored to the spatio-temporal constraints of the task and in consideration of the energetic costs.

Conclusion

We conclude that subjects modulate joint impedance, probably by making use of both intrinsic muscle properties and spinal reflexes, to meet accuracy demands during goal-directed movements, at least towards the end of movement. During less constrained, more natural tasks, impedance modulation is probably less apparent because other, less energy consuming, strategies are employed in combination with impedance modulation.

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